

Assignment 5 — Error Handling & Status Codes

Concept: Central error handler, throw vs res.send, 400/401/403/404/409

Level	Medium
Time	1.5 hours
File banani hai	src/utils/apiError.js , src/middlewares/errorHandler.js

Kya seekhoge

- Error ko **ek jagah** handle karna (har controller me try/catch nahi)
- Kaunsi galti pe kaunsa status code
- Mongoose ke raw errors ko insaan-friendly banana
- Production me stack trace kyun nahi bhejte

Part A — apiError.js banao

Ek file jisme error banane wale **simple functions** hon (class **nahi**):

```
const createError = (statusCode, message, errors = []) => {
  const error = new Error(message);
  error.statusCode = statusCode;
  error.errors = errors;
  return error;
};
```

Iske upar ye shortcuts banao: `badRequest()` , `unauthorized()` , `forbidden()` , `notFound()` , `conflict()`

Sawaal: `new Error()` kyun use kiya, plain object `{}` kyun nahi?

Part B — errorHandler.js banao

Do middleware export karo:

1. **notFound** — koi route match na ho to 404 JSON de
2. **errorHandler** — ye cases handle kare:

Error	Status	Message
Humara <code>createError</code> wala	jo diya ho	jo diya ho
<code>err.name === "CastError"</code>	400	<code>Invalid value for 'xyz'</code>
<code>err.name === "ValidationError"</code>	400	field-wise errors array
<code>err.code === 11000</code>	409	<code>This email already exists</code>
Baaki sab kuch	500	<code>Internal server error</code>

Response ka shape:

```
{ success: false, message: "...", errors: [...] }
```

`stack` sirf tab bhejo jab `NODE_ENV` production **na** ho.

Part C — Ye buggy file theek karo

```
// buggy.js – isme 5 galtiyan hain, sab dhoondho aur fix karo
const express = require("express");
const app = express();

app.use(errorHandler); // (1)
app.use("/reviews", reviewRouter);

const getReview = async (req, res) => { // (2)
  const review = await ReviewModel.findById(req.params.id);
  if (!review) {
    res.status(200).json({ success: false, message: "not found" }); // (3)
  }
  res.json(review); // (4)
};

app.use((err, req, res) => { // (5)
  res.status(500).send(err.message);
});
```

Har galti ke saath likho: **kya galat hai** aur **kya hoga agar theek na karein**.

Part D — Status code quiz

Har situation ke saamne sahi code likho:

#	Situation	Code?
1	Naya review ban gaya	
2	<code>rating: 15</code> bheja	
3	Token bheja hi nahi	
4	Token sahi hai par role <code>user</code> hai, delete admin-only hai	
5	<code>/getSingleReview/<id jo exist nahi karti></code>	
6	Wahi email se dobara register	
7	Search me kuch nahi mila	
8	Mongoose connection tut gaya	

Soch ke batao

- 401 aur 403** me fark ek line me. Kaunsa "login karne se theek ho jaayega"?
- Search me kuch na mile to `404` kyun galat hai?
- Ye kyun galat hai — 2 problem batao: `js res.status(200).json({ success: false, message: "Product not found" });`
- Error middleware me 4 parameter kyun likhte hain jab `next` use hi nahi hota?

Bonus

`errorHandler` me JWT ke 2 errors bhi handle karo: `JsonWebTokenError` aur `TokenExpiredError` — dono `401`, par **message** alag. Frontend ko alag message se kya fayda hoga?